2.1. Using hockey stick decomposition or otherwise find the sums

$$\sum_{k=1}^{n} 6k^2 + 4k.$$

2.2 Take a deep breath and calculate

$$\binom{-1}{4}, \quad \binom{4/3}{3}, \quad \binom{-1/2}{4}.$$

3.1. Using Gauss summation trick or otherwise find the sum

$$-3\binom{n}{0} - 1\binom{n}{1} + 1\cdot\binom{n}{2} + 3\cdot\binom{n}{3} + \cdots + (2n-3)\binom{n}{n}.$$

3.2. Consider the equation $a + b + c = 11$ with integers $a \geq 1$, $b \geq 2$, $c \geq 3$.

(a) How many solutions are there?

(b) Which solution would you encode as $\star\star \mid\mid \star\star\star$?

4.1. Solve the recurrence equation:

$$b_n = \frac{n+3}{n+4} b_{n-1}, \quad b_1 = 10.$$

4.2. Here L is the lag operator and $x_n = 5^n$. Find the values

$$(1-5L)x_n, \quad (1-2L)x_n, \quad (1-5L)(1-3L)x_n.$$

5.1. Consider the recurrence equation $y_n - 8y_{n-1} + 15y_{n-2} = 0$.

Find all solutions of the form $y_n = \lambda^n$.

5.2. You know that $r_n = c\cdot 2^n + d\cdot n\cdot 3^n + n$ is the solution of some recurrence equation with initial conditions $r_0 = 3$ and $r_1 = 10$.

Find c and d.

6.1. Calculate the following complex values:

$$(6-3i)\cdot(2+5i), \quad \frac{1-8i}{1+4i}, \quad (1-4i)^3.$$

6.2. Consider the recurrence equation

$$a_n - 4a_{n-1} + 5a_{n-2} = 4n - 14.$$

Find the solution of the form $a_n = \alpha n + \beta$.

7.1. Extract the sequence (a_n) from the generating functions:

$$g(t) = \frac{1}{1-5t}, \quad g(t) = \frac{1}{1+2t}, \quad g(t) = \frac{t}{1-3t}.$$

7.2. Decompose the fraction into a sum of more simple fractions

$$g(t) = \frac{5-5t}{(1+2t)(1-3t)}.$$